REAR WHEEL STUD REPLACEMENT

AFFECTED VEHICLES: **CERTAIN 1999 AND 2000 MODEL YEAR E-350 CUTAWAY AND STRIPPED CHASSIS VEHICLES EQUIPPED WITH SINGLE REAR WHEEL DANA 70 AXLE.**

OVERVIEW

This procedure is for the installation of ***new*** shorter wheel studs. Some vehicles may have the proper length studs already installed. Therefore, the studs on each vehicle must be measured to determine if ***new*** studs are required. Also, on vehicles which require the ***new*** studs, the rear wheel mounting holes will be inspected for damage. Replacement of the rear wheel studs on this axle requires removal and disassembly of the rear axle hubs. The rear wheel bearings will be repacked with grease and ***new*** hub seals installed. Some rear axle fluid loss will occur during this procedure which makes it necessary to check and top off the rear axle fluid.

INSPECTION

1. Raise and support the vehicle.
2. Remove the rear tire and wheel assemblies.
3. NOTE: Measure the length of the wheel studs from the hub flange to the end of the studs.

   If the vehicle's studs do not exceed 54 mm (2-1/8 in), the vehicle does not require the installation of ***new*** wheel studs.
   - Install the rear tire and wheel assemblies.
   - Lower the vehicle.
   - Tighten the lug nuts to 200 Nm (148 lb-ft), then return the vehicle to the owner.

   If the vehicle's studs exceed 54 mm (2-1/8 in), the vehicle requires the installation of ***new*** wheel studs. Continue with the procedure.

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4. Inspect the wheel mounting holes for gouges, enlarged or irregular shape. Replace damaged wheels. See Figure 1.

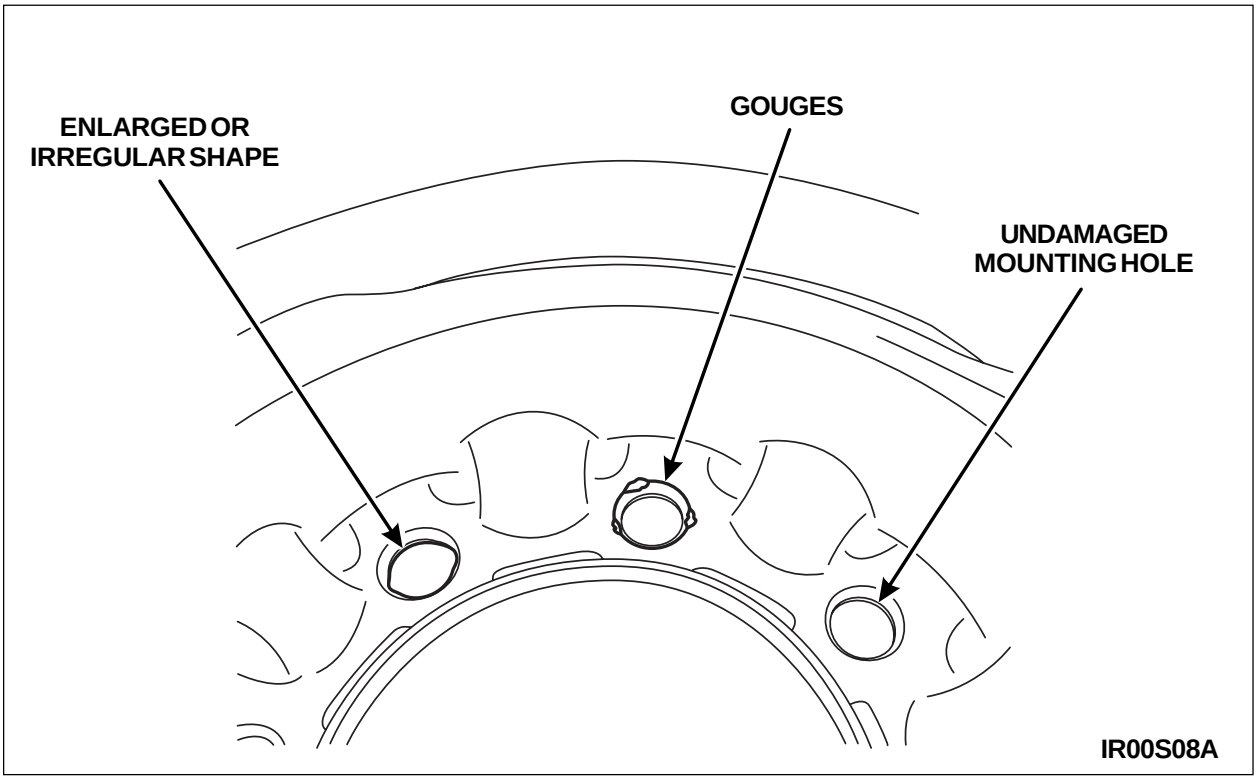


FIGURE 1

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WHEEL STUD REPLACEMENT

NOTE: Perform wheel stud replacement on both left and right side rear hubs.

1. **CAUTION: Never allow the rear disc brake caliper to hang from the brake hose. Provide suitable support.**

   Remove the brake caliper anchor plate and brake caliper as an assembly. Support anchor plate and caliper with mechanics wire. See Figure 2.

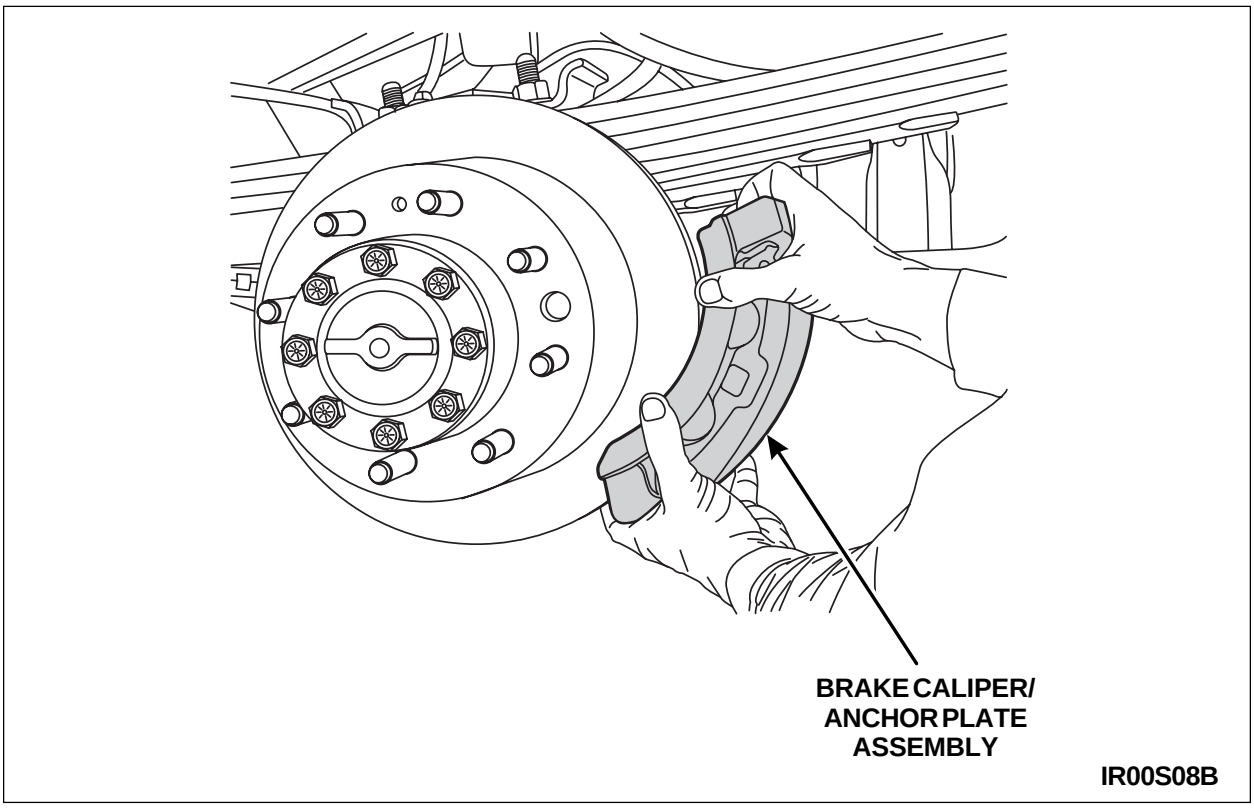


FIGURE 2

2. Position a drain pan under the rear hub.

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3. Remove the rear axle shaft bolts, then slide the axle shaft out of the hub. Discard the axle shaft gasket. See Figure 3.

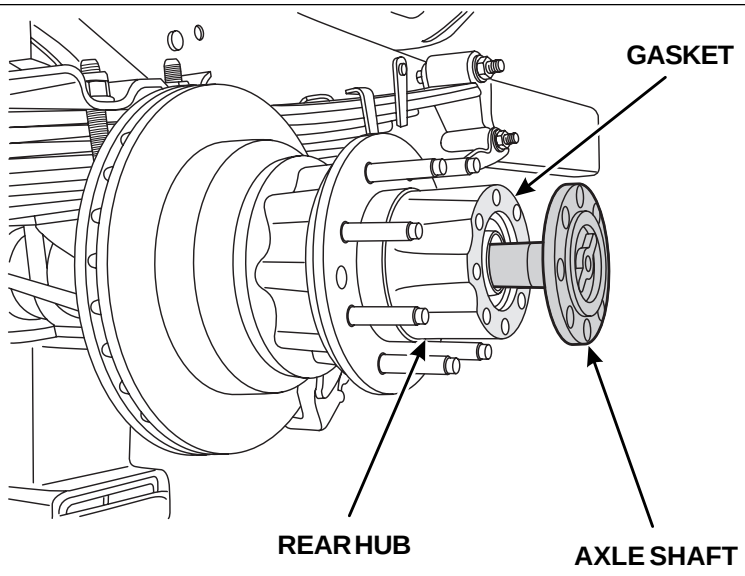


FIGURE 3

4. NOTE: Make sure that the drive tangs on the hub locknut wrench engage the four slots of the hub locknut.

   Remove the hub locknut using hub locknut wrench T88T-4252-A. See Figure 4.

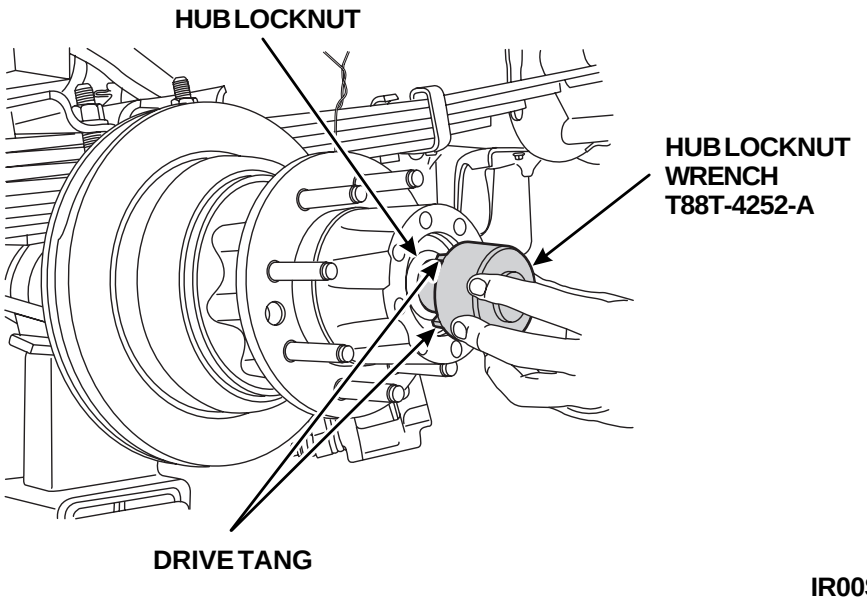


FIGURE 4

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5. Remove the outer wheel bearing, then remove the hub from the axle.
6. Mount the hub in a vise, then place a mark on the disc and the hub as a reference for reassembling. See Figure 5.
7. **CAUTION: The disc is attached to the hub with external Torx® bolts and requires an E18 external Torx® socket.**

   Separate the disc from the hub. See Figure 5.
8. NOTE: ***New*** rear hub seals must be installed when the rear hub is removed.

   Remove the rear hub seal and inner bearing. See Figure 5.

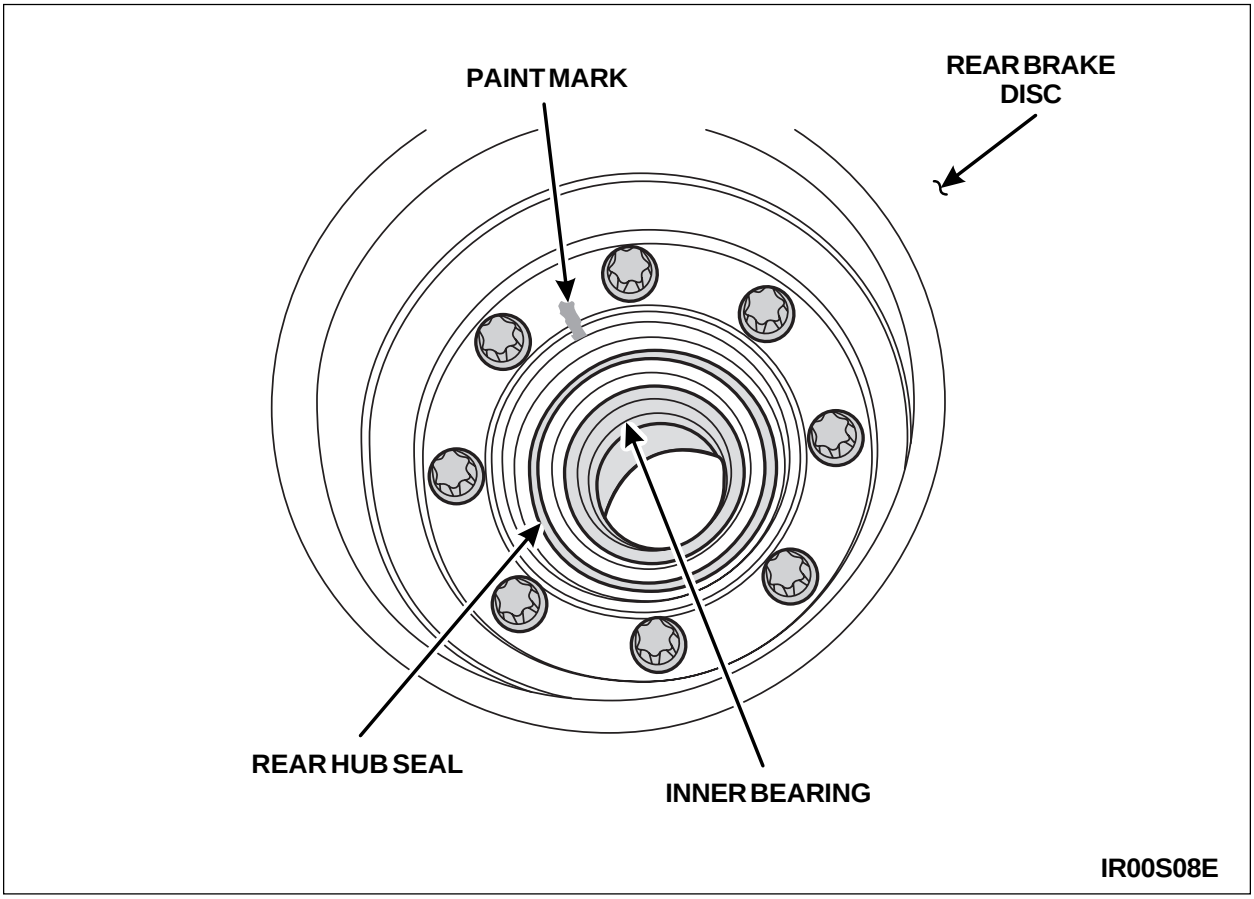


FIGURE 5

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9. Remove the wheel studs from the hub as follows. See Figure 6.
   - Position the hub into a press.
   - Support the hub flange with a suitable deep well socket.
   - Press the wheel stud out of the hub.
   - Reposition the rear hub and socket to remove the remaining wheel studs.

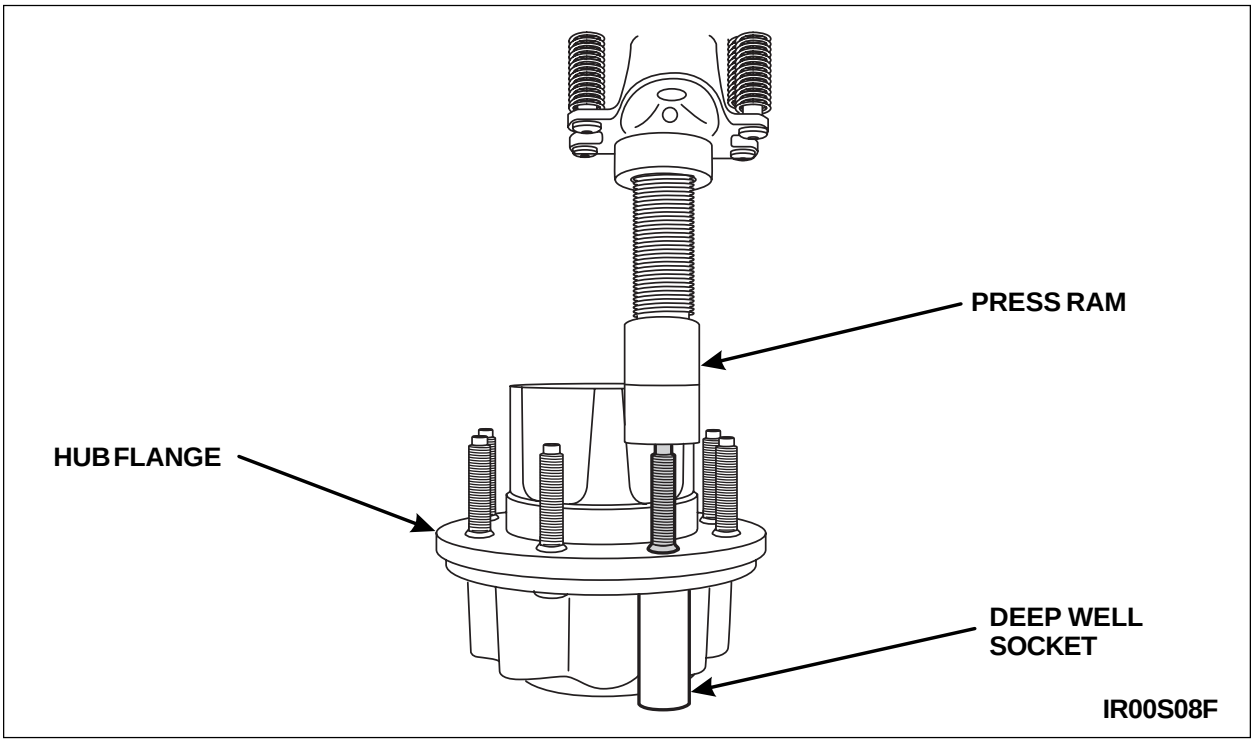


FIGURE 6

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10. Install ***new*** studs in the hub as follows. See Figure 7.
    - Reposition hub in press.
    - Support the hub flange with a suitable deep well socket.
    - Position and press the ***new*** wheel stud into the rear hub until the wheel stud head seats against the hub flange.
    - Reposition the hub and socket to install the remaining ***new*** wheel studs.

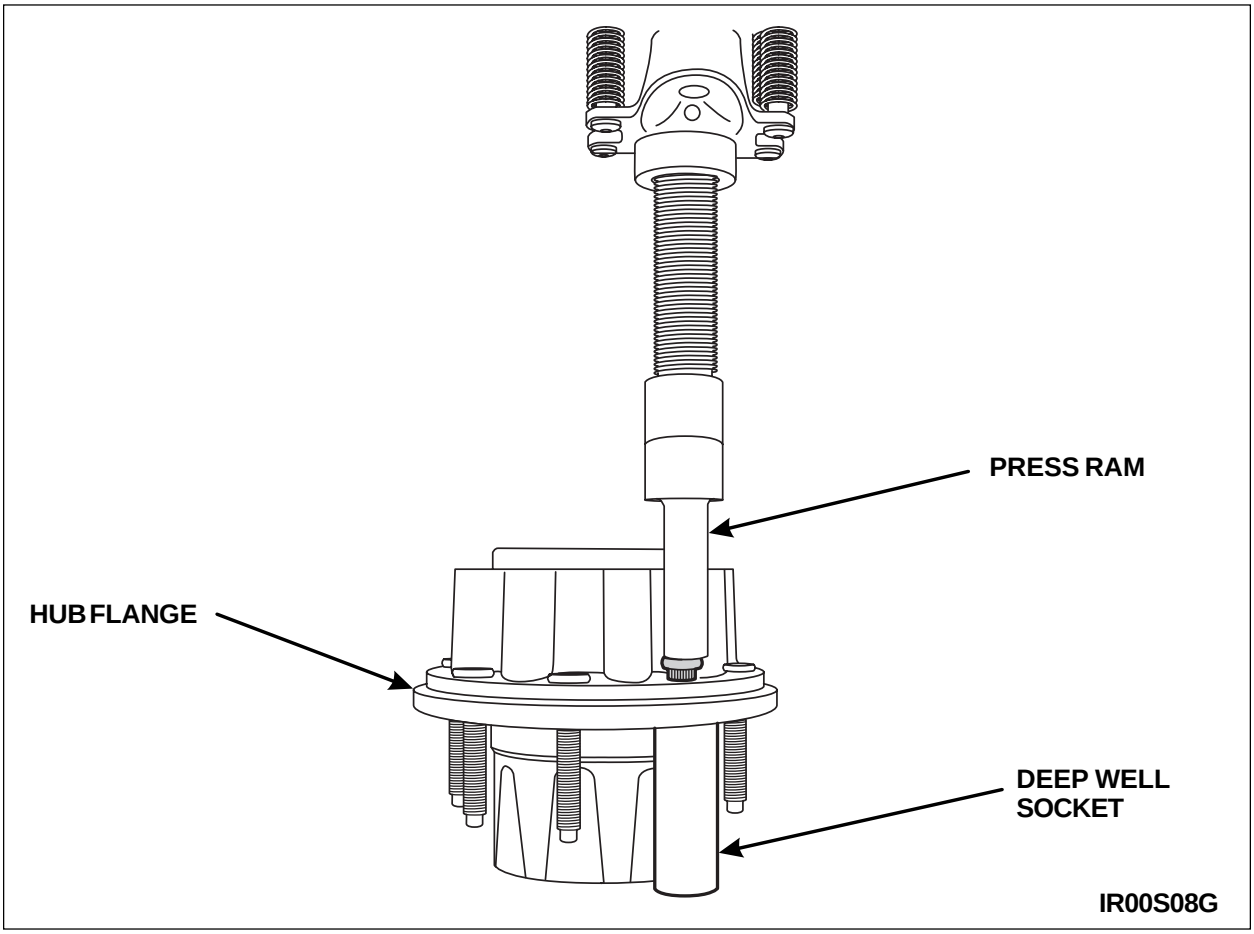


FIGURE 7

11. Clean any residual sealer from the seal mounting surface of the rear hub.
12. Pack the wheel bearings with Premium Long-Life Grease XG-1-C or equivalent.
13. Remount the hub in a vise, then install the original inner bearing and a ***new*** rear hub seal using a suitable seal installer.
14. NOTE: Align the marks on the rear brake disc and the hub.

    Attach the rear brake disc to the hub. Tighten bolts to 127 Nm (94 lb-ft).
15. Position the rear hub onto the rear axle.

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16. Install the outer bearing and hub locknut.
17. Tighten the hub locknut as follows.
    - Tighten to 95 Nm (70 lb-ft).
    - Loosen 90 degrees.
    - Tighten to 24 Nm (18 lb-ft).
18. NOTE: Do not final tighten the axle shaft bolts at this time.

    Position a ***new*** gasket onto the axle shaft, then install the axle shaft.
19. Install the brake caliper anchor plate and brake caliper assembly. Tighten bolts to 173 Nm (128 lb-ft).
20. Install the rear tire and wheel assemblies.
21. Lower the vehicle.
22. Tighten the lug nuts to 200 Nm (148 lb-ft).
23. Tighten rear axle shaft bolts to 138 Nm (102 lb-ft).
24. NOTE: Install one (1) ounce of Additive Friction Modifier C8AZ-19B546-A to axles equipped with limited-slip differential assemblies.

    Check and top off the rear axle with SAE 80W-90 Premium Rear Axle Lubricant XY-80W90-QL.

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